

Mathematics

Stage 9

Paper 1

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

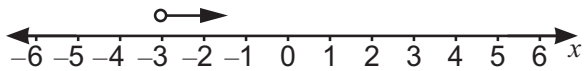
Plotting points

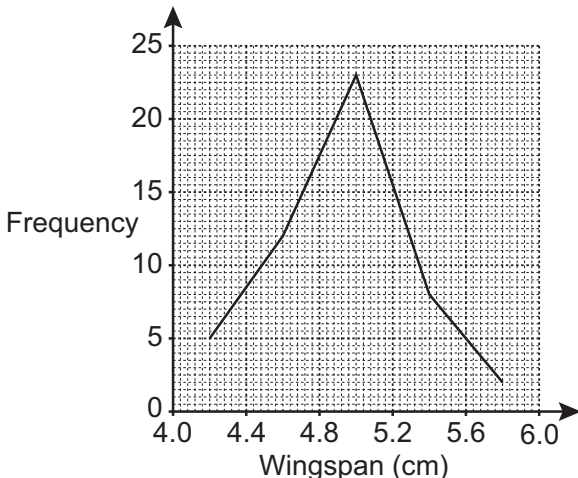
The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	1 ☐ 2 ☒ 3 ☐ 4 ☐	1		Accept any clear indication.
2(a)	0.6	1		Accept equivalent fractions or percentage, e.g. $\frac{6}{10}$ or 60%
2(b)	0.3	1		Accept if not inserted in table, but clearly the final answer. Accept equivalent fractions or percentage, e.g. $\frac{3}{10}$ or 30%
3	(x =) 4	1		
4	Correct equilateral triangle with correct construction arcs left visible.	1		Label C not required. Tolerance ± 2 mm
5	Independent ☒ ☐ ☒ Not independent ☐ ☒ ☐	1		Accept any clear indication. All three answers correct for the mark.
6	(4, 6)	1		
7	6 ☐ 9 ☐ 18 ☐ 28 ☒ 81 ☐	1		Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance									
8	7	1											
9	72 (cm ²)	1											
10	$\frac{9}{10}$	2	Award 1 mark for $\frac{2}{5} \times \frac{9}{4}$ or $\frac{18}{45} \div \frac{20}{45}$	1 mark implied by any equivalent fraction to $\frac{9}{10}$ e.g. $\frac{18}{20}$									
11(a)	(Stage 3) She travels at (a constant speed of) 50 km/h for $\frac{1}{2}$ an hour. (Stage 4) She travels at (a constant speed of) 25 km/h for $\frac{1}{2}$ an hour.	2	Award 1 mark for one stage correctly described or for two correct speeds with times missing/wrong.	Accept equivalent times, e.g. 30 minutes.									
11(b)	A straight line from (3 pm, 87.5) to (4.45 pm, 0).	1		Accept values closer to 4.45 pm than 4.30 pm or 5 pm.									
12	Five letters (B to F) correctly placed in the Carroll diagram. <table><tr><td></td><td>Congruent to shape X</td><td>Not congruent to shape X</td></tr><tr><td>Similar to shape X</td><td>(A) D</td><td>B F</td></tr><tr><td>Not similar to shape X</td><td></td><td>C E</td></tr></table>		Congruent to shape X	Not congruent to shape X	Similar to shape X	(A) D	B F	Not similar to shape X		C E	2	Award 1 mark for three or four letters (B to F) correctly placed in the Carroll diagram.	
	Congruent to shape X	Not congruent to shape X											
Similar to shape X	(A) D	B F											
Not similar to shape X		C E											
13	8	1											

Question	Answer	Marks	Part Marks	Guidance
14(a)	7.6×10^7 76×10^7 (7.6×10^8) 7.6×10^9	1		Accept any clear indication.
14(b)	$4.6 \times 10^{-5}(\text{m})$	1		
15	0.84 (km)	1		
16	Any correct example of a fraction with an odd denominator showing it is not a recurring decimal, e.g. $\frac{1}{5} = 0.2$	1		Must show fraction and correct decimal equivalent for the mark.
17(a)	16	1		
17(b)	0 and 6	2	Award 1 mark for one correct value.	Accept 0 and 6 in either order for 2 marks.
18(a)	$x > -3$	2	Award 1 mark for gathering the terms in x on one side and constant terms on the other side, e.g. $2x - 6x < 3 - -9$ (ignore inequality for this mark, may replace with =) or Award 1 mark for $x > -3$ in the working with just -3 on the answer line or for $x = -3$ on the answer line.	
18(b)		1	Follow through <i>their</i> inequality from part (a).	

Question	Answer	Marks	Part Marks	Guidance
19	$2\frac{1}{15}$ correct answer only	3	Award 2 marks for $\frac{31}{15}$ or equivalent fraction. or Award 1 mark for $\frac{22}{5}$ or $\frac{8}{3}$ or $2 + \frac{2}{5} - \frac{1}{3}$ or $4\frac{2}{5} - 2\frac{1}{3}$ or $4\frac{6}{15} - 2\frac{10}{15} + \frac{5}{15}$	
20	Straight lines joining (4.2, 5) and (4.6, 12) and (5.0, 23) and (5.4, 8) and (5.8, 2). 	3	Award 1 mark for four or five plots correct horizontally ($x = 4.2, 4.6, 5.0, 5.4, 5.8$). and Award 1 mark for four or five plots correct vertically (frequency = 5, 12, 23, 8, 2).	Mark intention.

Question	Answer	Marks	Part Marks	Guidance
21	Translation of $\begin{pmatrix} -4 \\ -9 \end{pmatrix}$	2	Award 1 mark for the word translation or for $\begin{pmatrix} -4 \\ -9 \end{pmatrix}$ or for a correct reflection of A in $y = x$ drawn on the grid.	Do not accept $\begin{pmatrix} -4 \\ -9 \end{pmatrix}$ expressed in words.
22	55(°)	2	Award 1 mark for 60(°) correctly marked at <i>BED</i> or 65(°) correctly marked at <i>BDE</i> or 65(°) correctly marked at <i>BAC</i> .	
23	Any pair of coordinates (a, b) , (c, d) where $\frac{a+c}{2} = 3.5$ and $\frac{b+d}{2} = -2$ but not $(3.5, -2)$ and $(3.5, -2)$ as this is not a line segment.	2	Award 1 mark for any pair of coordinates (a, b) , (c, d) where $\frac{a+c}{2} = 3.5$ or $\frac{b+d}{2} = -2$	Accept a and c as 3.5 or b and d as -2 , but not both, for 1 or 2 marks, as appropriate. 0 marks for $(3.5, -2)$ and $(3.5, -2)$.
24	$1 - 3q$ correct answer only	1		
25(a)	$2n + 5 = 91$ $2n = 86$ $n = 43$ or 86 is even/a multiple of 2	1		Full working required for the mark. Accept sequence extended up to 91
25(b)	$3n - 6 = 91$ $3n = 97$ $n = 32.3 \dots$ or equivalent or 97 is not a multiple of 3	1		Full working required for the mark. Accept sequence extended up to 93
25(c)	27	2	Award 1 mark for $2n + 5 = 3n - 6$ implied by $n = 11$	

Question	Answer	Marks	Part Marks	Guidance
26	Boys ticked and more boys have (2 or 3 or) 4 or more siblings or equivalent and Boys ticked and the range for boys is 4 or more and/or the range for girls is 3	2	Award 1 mark for Boys ticked and more boys have (2 or 3 or) 4 or more siblings or equivalent or Boys ticked and the range for boys is 4 or more and/or the range for girls is 3	Or equivalent for first explanation, e.g. • Fewer girls have (2 or 3 or) 4 or more siblings. • No girls have 4 or more siblings. • More girls have 0 or 1 sibling. • Fewer boys have 0 or 1 sibling. For first explanation, accept reference to taller bars, e.g. the bar for boys is taller than for girls for (2 or 3 or) 4 or more siblings.
27	$(x =) 0.5$ and $(y =) 3$	3	Award 2 marks for $x = 0.5$ or $y = 3$ or Award 1 mark for a correct method for eliminating either x or y , e.g. • Re-arranging one of the equations to make one variable the subject and then substitute <i>their</i> arrangement into the other equation. • Making the coefficients of x or y equal with no more than one arithmetic error or sign error, followed by an appropriate, consistent subtraction or addition across all three terms. • Correct substitution and evaluation from incorrect first value, i.e. two values satisfying one of the original equations.	Accept $x = \frac{1}{2}$